Running the week 4 Flutter apps in VS Code

Project: `code/week04/flutter_examples`

What you need

- Flutter, and VS Code with the **Flutter** extension (install guide: `001_install_vscode`).
- A device: **Chrome** or **Windows (desktop)** is enough this week; no Android emulator needed.

```
flutter doctor   # tools and devices
```

1. Get the code

```
git pull         # already cloned
git clone https://github.com/maoude/mobile_pro_2026↩
    .git
```

File → **Open Folder...** and open `code/week04/flutter_examples` *itself*, not a parent folder: that is how VS Code recognises a Flutter project.

2. Prepare the project (once)

Terminal → **New Terminal**, then:

```
flutter pub get   # download the packages
flutter create .  # add android/, web/, windows/...
```

`flutter create .` adds only the missing platform folders; it does not change your files.

3. Choose a device

Bottom right of the VS Code window: click the device name (maybe **No Device**) and pick **Chrome**, **Windows (desktop)** or an emulator.

4. Run one app

Every file of `lib` is a complete app with its own `main`.

1. Open a file, e.g. `lib/05_text_style.dart`.
2. Click **Run** in the **Run | Debug** line above `void main()`.

```
flutter run -t lib/05_text_style.dart
```

`lib/main.dart` starts the last step of the lecture (08).

While the app runs

Keys	Effect
`Ctrl+S`	**hot reload**: save, the app updates in a second
`Ctrl+Shift+F5`	**hot restart**: start again from the beginning
`Shift+F5`	stop the app

Automated checks

```
flutter test     # tests in the folder test/
```

Each test starts one step and checks the screen. Expected last line: `All tests passed!`

What you should see: the lecture steps

File	On the screen
00	"Hello, Rami" and a checkbox; tick it: "Accepted"
01	a completely white screen
02	"Hello World" in red with a yellow underline (expected)
03	normal small dark text: the `Scaffold` fixed it
04	a top bar with the centered title "Home Page"
05	larger, deep purple "Hello World"
06	"Text 1" in purple, from `MyTextWidget`
07	"Text 1" and "Text 2", one below the other
08	plus "Text 3", "Text 4", "Text 5" spread evenly
09	screen size, pixel ratio, physical size (depends on device)

Going further and Part 2

File	On the screen
10	teal bar: menu icon left, search and settings right; a tap shows a message at the bottom
11	light page, a `Card` in the middle, a bottom bar
12	grey fixed strip, then three areas in the ratio 1:2:1, also when the window is resized
13	scrollable page: text, three buttons, logo, red heart, "Last button: none"; a Row and a `Stack`
14	a square "Taps: 0": tap $+1$, long press resets
15	two texts and a tree button: the tree is printed in the terminal or Debug Console

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Example 13 shows an image from `assets/`, already declared in `pubspec.yaml`: nothing to set up.

Try it yourself (save after each change)

- `MyTextWidget`: change `fontSize` and `color`.
- Add a sixth text to the Row.
- `spaceEvenly` → `spaceAround`, `spaceBetween`; compare.
- A `SizedBox` height: 16.0 → 40.0.
- 12: `flex` 1, 1, 3, then a fourth `Expanded`.
- 14: an `onDoubleTap` that adds 10.
- 15: a third `Text`, then find it in the dump.

When something goes wrong

No **Run | Debug** line	install the Flutter extension; open `flutter_examples` itself
No Device	choose Chrome; check `flutter doctor`
No platform configured	run `flutter create .` once
Step 2 looks ugly	on purpose; step 3 fixes it
Waiting for . . . startup lock	another Flutter command runs; wait or close it

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